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Assignment 1 on Newton-Rhapson algorithm

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Advanced Astrodynamics

ABSTRACT

This report analyzes the convergence behavior of three numerical methods for solving Kepler's equation, namely the Newton–Raphson, Fixed-Point Iteration, and Halley's methods. Each method is tested using three different initial guesses, and their convergence properties are assessed over a wide range of eccentricities and mean anomalies.

1 INTRODUCTION

Kepler's equation is a fundamental relation in celestial mechanics, connecting the mean anomaly M, the eccentric anomaly E, and the eccentricity e through:

$$M = E - e\sin E. \tag{1}$$

Since this equation cannot be solved analytically for E, iterative numerical methods are required.

2 NUMERICAL METHODS

Three iterative methods were implemented and analyzed: Newton–Raphson, Fixed-Point, and Halley's method. Each is briefly summarized below.

2.1 Fixed-Point Iteration

A simpler approach consists of rearranging Kepler's equation as:

$$E = M + e\sin E, \tag{2}$$

and using the iteration:

$$E_{n+1} = M + e\sin E_n. \tag{3}$$

Although easy to implement, convergence is guaranteed only when $e < 0.66$, and the rate is linear.

2.2 Newton–Raphson Method

The Newton–Raphson (NR) method is a second-order iterative technique that updates the solution as:

$$E_{n+1} = E_n - \frac{E_n - e\sin E_n - M}{1 - e\cos E_n}. \tag{4}$$

The method generally converges quadratically near the solution if the initial guess is close enough.

2.3 Halley's Method

Halley's method extends Newton–Raphson to third-order convergence. The iteration formula is:

$$E_{n+1} = E_n - \frac{f(E_n)}{f'(E_n) - \frac{1}{2} f(E_n) f''(E_n)/f'(E_n)}, \tag{5}$$

where

$$f(E) = E - e\sin E - M, \quad f'(E) = 1 - e\cos E, \quad f''(E) = e\sin E.$$

This method typically converges faster than NR, particularly for high eccentricities.

3 METHODOLOGY

The analysis was performed using Python, testing each solver over a grid of mean anomalies $M \in [0, 2\pi]$ and eccentricities $e \in [0, 1)$. The number of iterations required to achieve a tolerance of 10^{-12} was recorded.

Three different initial guesses were evaluated:

1. $E_0 = M$
2. $E_0 = \pi$
3. $E_0 = M + e\cos M$

4 RESULTS AND DISCUSSION

Figures **??**–1 show the convergence maps and iteration histories. The Newton–Raphson and Halley's methods converge rapidly for all eccentricities except near $e \approx 1$ and $M \approx 0$. The fixed-point iteration diverges for large e, confirming its limited applicability.

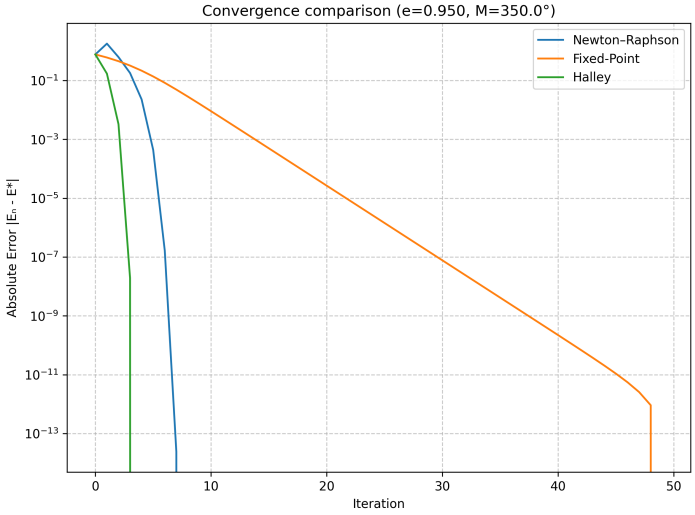
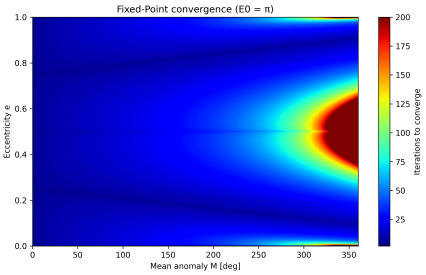


Figure 1. Comparison of convergence rates among the three methods.

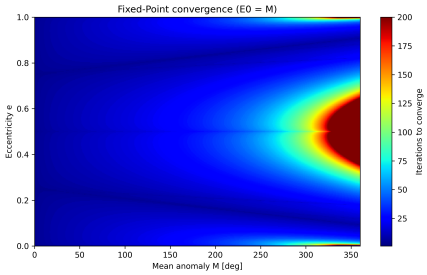
5 CONCLUSION

The Newton–Raphson method offers an excellent balance of simplicity and performance. Halley's method converges faster but requires more computation per iteration. The fixed-point approach should only be used for small eccentricities.

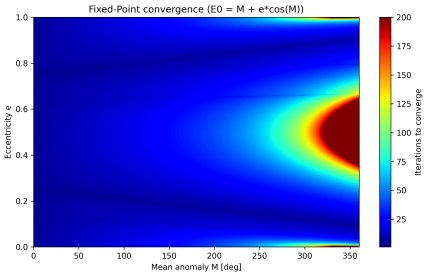
A GRAPHICAL RESULTS



(a) Convergence map for fixed-point method using $E_0 = \pi$.

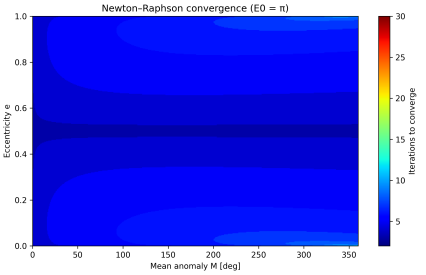


(b) Convergence map for fixed-point method using $E_0 = M$.

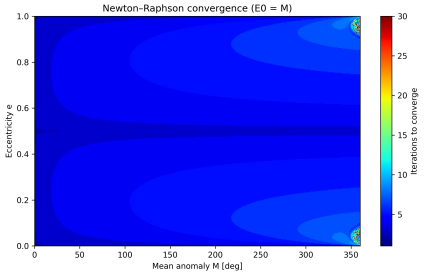


(c) Convergence map for fixed-point method using $E_0 = M + e\cos(M)$.

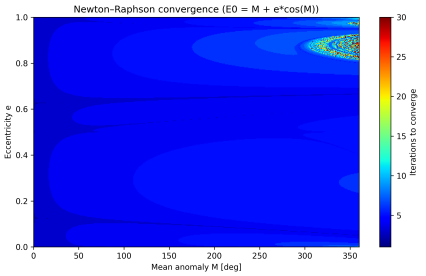
Figure 2. Convergence maps for the fixed-point method using different initial guesses E_0.



(a) Convergence map for Newton–Raphson method using $E_0 = \pi$.

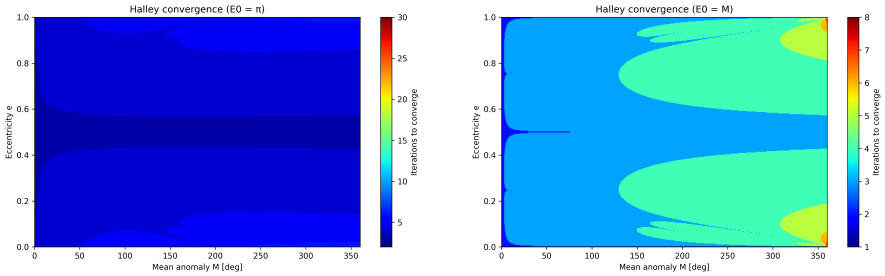


(b) Convergence map for Newton–Raphson method using $E_0 = M$.



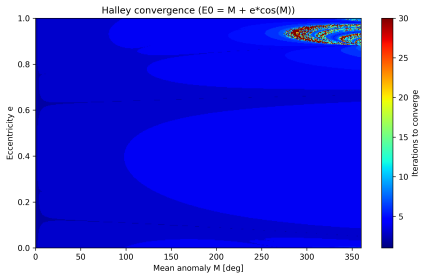
(c) Convergence map for Newton–Raphson method using $E_0 = M + e\cos(M)$.

Figure 3. Convergence maps for the Newton–Raphson method using different initial guesses E_0.



(a) Convergence map for Halley method using $E_0 = \pi$.

(b) Convergence map for Halley method using $E_0 = M$.



(c) Convergence map for Halley method using $E_0 = M + e\cos(M)$.

Figure 4. Convergence maps for the Halley method using different initial guesses E_0.